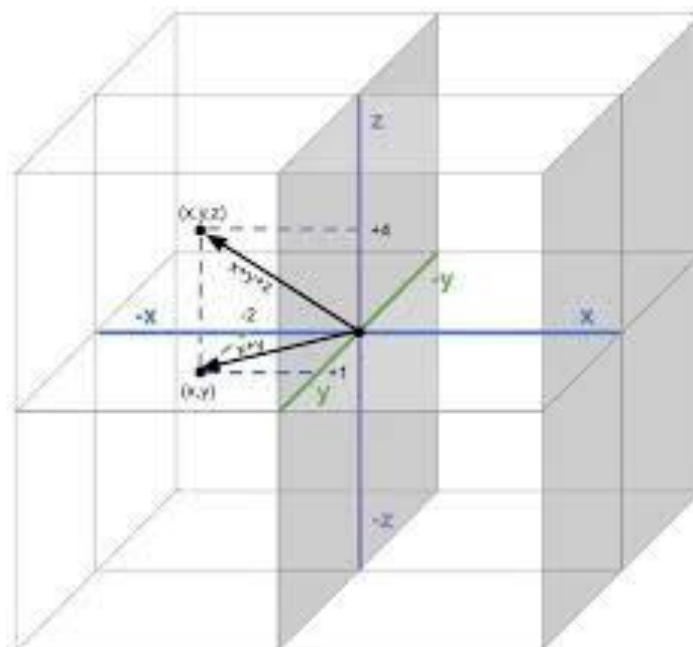


Linear Algebra

Math 1057

Instructor

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Lecture 19

Change of Basis

19.1 Review of Coordinate Mappings on $\mathbb{R}^n$

Let $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis for $\mathbb{R}^n$ and let

$$\mathbf{P}_{\mathcal{B}} = [\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n].$$

If $\mathbf{x} \in \mathbb{R}^n$ and $[\mathbf{x}]_{\mathcal{B}}$ is the coordinate vector of $\mathbf{x}$ in the basis $\mathcal{B}$ then

$$\mathbf{x} = \mathbf{P}_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}.$$

The components of the vector $\mathbf{x}$ are the coordinates of $\mathbf{x}$ in the standard basis $\mathcal{E} = \{\mathbf{e}_1, \dots, \mathbf{e}_n\}$. In other words,

$$[\mathbf{x}]_{\mathcal{E}} = \mathbf{x}.$$

Therefore,

$$[\mathbf{x}]_{\mathcal{E}} = \mathbf{P}_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}.$$

We can therefore interpret $\mathbf{P}_{\mathcal{B}}$ as the matrix mapping that maps the $\mathcal{B}$ -coordinates of $\mathbf{x}$ to the $\mathcal{E}$ -coordinates of $\mathbf{x}$. To make this more explicit, we sometimes use the notation

$${}_{\mathcal{E}}\mathbf{P}_{\mathcal{B}}$$

to indicate that ${}_{\mathcal{E}}\mathbf{P}_{\mathcal{B}}$ maps $\mathcal{B}$ -coordinates to $\mathcal{E}$ -coordinates:

$$[\mathbf{x}]_{\mathcal{E}} = ({}_{\mathcal{E}}\mathbf{P}_{\mathcal{B}})[\mathbf{x}]_{\mathcal{B}}.$$

If we multiply the equation

$$[\mathbf{x}]_{\mathcal{E}} = ({}_{\mathcal{E}}\mathbf{P}_{\mathcal{B}})[\mathbf{x}]_{\mathcal{B}}$$

on the left by the inverse of ${}_{\mathcal{E}}\mathbf{P}_{\mathcal{B}}$ we obtain

$$({}_{\mathcal{E}}\mathbf{P}_{\mathcal{B}})^{-1}[\mathbf{x}]_{\mathcal{E}} = [\mathbf{x}]_{\mathcal{B}}$$

Hence, the matrix $({}_{\mathcal{E}}\mathbf{P}_{\mathcal{B}})^{-1}$ maps standard coordinates to $\mathcal{B}$ -coordinates, see Figure 19.1. It is natural then to introduce the notation

$${}_{\mathcal{B}}\mathbf{P}_{\mathcal{E}} = ({}_{\mathcal{E}}\mathbf{P}_{\mathcal{B}})^{-1}$$

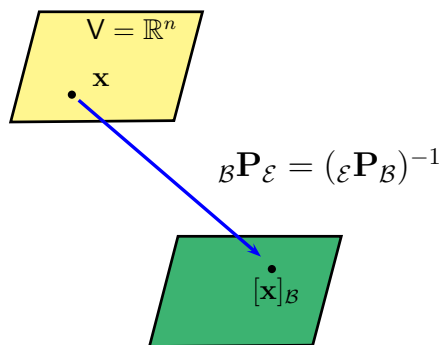


Figure 19.1: The matrix ${}_B \mathbf{P}_E$ maps $\mathcal{E}$ coordinates to $\mathcal{B}$ coordinates.

Example 19.1. Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} -8 \\ 2 \\ 3 \end{bmatrix}.$$

- Show that the set of vectors $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ forms a basis for $\mathbb{R}^n$.
- Find the change-of-coordinates matrix from $\mathcal{B}$ to standard coordinates.
- Find the coordinate vector $[\mathbf{x}]_{\mathcal{B}}$ for the given $\mathbf{x}$.

Solution. Let

$$\mathbf{P}_{\mathcal{B}} = \begin{bmatrix} 1 & -3 & 3 \\ 0 & 4 & -6 \\ 0 & 0 & 3 \end{bmatrix}$$

It is clear that $\det(\mathbf{P}_{\mathcal{B}}) = 12$, and therefore $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly independent. Therefore, $\mathcal{B}$ is a basis for $\mathbb{R}^n$. The matrix $\mathbf{P}_{\mathcal{B}}$ takes $\mathcal{B}$ -coordinates to standard coordinates. The $\mathcal{B}$ -coordinate vector $[\mathbf{x}]_{\mathcal{B}} = (c_1, c_2, c_3)$ is the unique solution to the linear system

$$\mathbf{x} = \mathbf{P}_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}$$

Solving the linear system with augmented matrix $[\mathbf{P}_{\mathcal{B}} \ \mathbf{x}]$ we obtain

$$[\mathbf{x}]_{\mathcal{B}} = (-5, 2, 1)$$

We verify that $[\mathbf{x}]_{\mathcal{B}} = (-5, 2, 1)$ are indeed the coordinates of $\mathbf{x} = (-8, 2, 3)$ in the basis

$$\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}:$$

$$\begin{aligned} (-5)\mathbf{v}_1 + (2)\mathbf{v}_2 + (1)\mathbf{v}_3 &= -5 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix} + \begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} -5 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -6 \\ 8 \\ 0 \end{bmatrix} + \begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix} \\ &= \underbrace{\begin{bmatrix} -8 \\ 2 \\ 3 \end{bmatrix}}_{\mathbf{x}} \end{aligned}$$

□

19.2 Change of Basis

We saw in the previous section that the matrix

$${}_{\mathcal{E}}\mathbf{P}_{\mathcal{B}}$$

takes as input the $\mathcal{B}$ -coordinates $[\mathbf{x}]_{\mathcal{B}}$ of a vector $\mathbf{x}$ and returns the coordinates of $\mathbf{x}$ in the standard basis. We now consider the situation of dealing with two basis $\mathcal{B}$ and $\mathcal{C}$ where neither is assumed to be the standard basis $\mathcal{E}$. Hence let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ and let $\mathcal{C} = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ be two basis of $\mathbb{R}^n$ and let

$${}_{\mathcal{E}}\mathbf{P}_{\mathcal{B}} = [\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n]$$

$${}_{\mathcal{E}}\mathbf{P}_{\mathcal{C}} = [\mathbf{w}_1 \ \mathbf{w}_2 \ \cdots \ \mathbf{w}_n].$$

Then if $[\mathbf{x}]_{\mathcal{C}}$ is the coordinate vector of $\mathbf{x}$ in the basis $\mathcal{C}$ then

$$\mathbf{x} = ({}_{\mathcal{E}}\mathbf{P}_{\mathcal{C}})[\mathbf{x}]_{\mathcal{C}}.$$

How do we transform $\mathcal{B}$ -coordinates of $\mathbf{x}$ to $\mathcal{C}$ -coordinates of $\mathbf{x}$, and vice-versa? To answer this question, start from the relations

$$\mathbf{x} = ({}_{\mathcal{E}}\mathbf{P}_{\mathcal{B}})[\mathbf{x}]_{\mathcal{B}}$$

$$\mathbf{x} = ({}_{\mathcal{E}}\mathbf{P}_{\mathcal{C}})[\mathbf{x}]_{\mathcal{C}}.$$

Then

$$({}_{\mathcal{E}}\mathbf{P}_{\mathcal{C}})[\mathbf{x}]_{\mathcal{C}} = ({}_{\mathcal{E}}\mathbf{P}_{\mathcal{B}})[\mathbf{x}]_{\mathcal{B}}$$

and because ${}_{\mathcal{E}}\mathbf{P}_{\mathcal{C}}$ is invertible we have that

$$[\mathbf{x}]_{\mathcal{C}} = ({}_{\mathcal{E}}\mathbf{P}_{\mathcal{C}})^{-1}({}_{\mathcal{E}}\mathbf{P}_{\mathcal{B}})[\mathbf{x}]_{\mathcal{B}}.$$

Hence, the matrix $(\varepsilon \mathbf{P}_C)^{-1}(\varepsilon \mathbf{P}_B)$ maps the $\mathcal{B}$ -coordinates of $\mathbf{x}$ to the $\mathcal{C}$ -coordinates of $\mathbf{x}$. For this reason, it is natural to use the notation (see Figure 19.2)

$${}_C \mathbf{P}_B = (\varepsilon \mathbf{P}_C)^{-1}(\varepsilon \mathbf{P}_B).$$

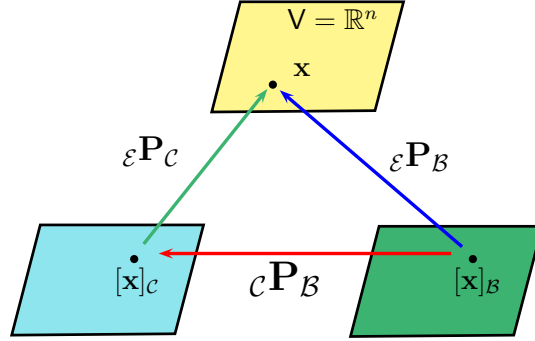


Figure 19.2: The matrix ${}_C \mathbf{P}_B$ maps $\mathcal{B}$ -coordinates to $\mathcal{C}$ -coordinates.

If we expand $(\varepsilon \mathbf{P}_C)^{-1}(\varepsilon \mathbf{P}_B)$ we obtain that

$$(\varepsilon \mathbf{P}_C)^{-1}(\varepsilon \mathbf{P}_B) = [(\varepsilon \mathbf{P}_C)^{-1} \mathbf{v}_1 \quad (\varepsilon \mathbf{P}_C)^{-1} \mathbf{v}_2 \quad \cdots \quad (\varepsilon \mathbf{P}_C)^{-1} \mathbf{v}_n].$$

Therefore, the i th column of $(\varepsilon \mathbf{P}_C)^{-1}(\varepsilon \mathbf{P}_B)$, namely

$$(\varepsilon \mathbf{P}_C)^{-1} \mathbf{v}_i,$$

is the coordinate vector of $\mathbf{v}_i$ in the basis $\mathcal{C} = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n\}$. To compute ${}_C \mathbf{P}_B$ we augment $\varepsilon \mathbf{P}_C$ and $\varepsilon \mathbf{P}_B$ and row reduce fully:

$$[\varepsilon \mathbf{P}_C \quad \varepsilon \mathbf{P}_B] \sim [\mathbf{I}_n \quad {}_C \mathbf{P}_B].$$

Example 19.2. Let

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -3 \end{bmatrix}, \begin{bmatrix} -2 \\ 4 \end{bmatrix} \right\}, \quad \mathcal{C} = \left\{ \begin{bmatrix} -7 \\ 9 \end{bmatrix}, \begin{bmatrix} -5 \\ 7 \end{bmatrix} \right\}$$

It can be verified that $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ and $\mathcal{C} = \{\mathbf{w}_1, \mathbf{w}_2\}$ are bases for $\mathbb{R}^2$.

- Find the matrix that takes $\mathcal{B}$ -coordinates to $\mathcal{C}$ -coordinates.
- Find the matrix that takes $\mathcal{C}$ -coordinates to $\mathcal{B}$ -coordinates.
- Let $\mathbf{x} = (0, -2)$. Find $[\mathbf{x}]_B$ and $[\mathbf{x}]_C$.

Solution. The matrix $\varepsilon \mathbf{P}_B = [\mathbf{v}_1 \quad \mathbf{v}_2]$ maps $\mathcal{B}$ -coordinates to standard $\mathcal{E}$ -coordinates. The matrix $\varepsilon \mathbf{P}_C = [\mathbf{w}_1 \quad \mathbf{w}_2]$ maps $\mathcal{C}$ -coordinates to standard $\mathcal{E}$ -coordinates. As we just showed, the matrix that maps $\mathcal{B}$ -coordinates to $\mathcal{C}$ -coordinates is

$${}_C \mathbf{P}_B = (\varepsilon \mathbf{P}_C)^{-1}(\varepsilon \mathbf{P}_B)$$

It is straightforward to compute that

$$({}_\varepsilon \mathbf{P}_C)^{-1} = \begin{bmatrix} -7/4 & -5/4 \\ 9/4 & 7/4 \end{bmatrix}$$

Therefore,

$${}_c \mathbf{P}_B = ({}_c \mathbf{P}_C)^{-1} ({}_c \mathbf{P}_B) = \begin{bmatrix} -7/4 & -5/4 \\ 9/4 & 7/4 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & -3/2 \\ -3 & 5/2 \end{bmatrix}$$

To compute ${}_B \mathbf{P}_C$, we can simply invert ${}_c \mathbf{P}_B$. One finds that

$$({}_c \mathbf{P}_B)^{-1} = \begin{bmatrix} 5 & 3 \\ 6 & 4 \end{bmatrix}$$

and therefore

$${}_B \mathbf{P}_C = \begin{bmatrix} 5 & 3 \\ 6 & 4 \end{bmatrix}$$

Given that $\mathbf{x} = (0, -2)$, to find $[\mathbf{x}]_B$ we must solve the linear system

$${}_c \mathbf{P}_B [\mathbf{x}]_B = \mathbf{x}$$

Row reducing the augmented matrix $[{}_c \mathbf{P}_B \ \mathbf{x}]$ we obtain

$$[\mathbf{x}]_B = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Next, to find $[\mathbf{x}]_C$ we can solve the linear system

$${}_c \mathbf{P}_C [\mathbf{x}]_C = \mathbf{x}$$

Alternatively, since we now know $[\mathbf{x}]_B$ and ${}_c \mathbf{P}_B$ has been computed, to find $[\mathbf{x}]_C$ we simply multiply ${}_c \mathbf{P}_B$ by $[\mathbf{x}]_B$:

$$[\mathbf{x}]_C = {}_c \mathbf{P}_B [\mathbf{x}]_B = \begin{bmatrix} 2 & -3/2 \\ -3 & 5/2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 5/2 \\ -7/2 \end{bmatrix}$$

Let's verify that $[\mathbf{x}]_C = \begin{bmatrix} 5/2 \\ -7/2 \end{bmatrix}$ are indeed the $\mathcal{C}$ -coordinates of $\mathbf{x} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$:

$${}_c \mathbf{P}_C [\mathbf{x}]_C = \begin{bmatrix} -7 & -5 \\ 9 & 7 \end{bmatrix} \begin{bmatrix} 5/2 \\ -7/2 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}.$$

□

After this lecture you should know the following:

- how to compute a change of basis matrix
- and how to use the change of basis matrix to map one set of coordinates into another